Project 4 Proposal

CS 528 Computer Security

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Overview

We propose a web-based password manager inspired by Avast Password Manager. It will replicate the core password vault and password generation functionality, as well as its user interface.

Planned Features

Our application will provide a password-protected vault for managing login credentials, payment cards, addresses, and secure notes. Users can add, edit, delete, search, and organize stored information, including favorites. The password generator will support configurable length, character sets, password regeneration, copy-to-clipboard, and strength feedback.

Implementation and Security

We plan to use JavaScript (React), HTML, CSS, and Vite for the application. We will encrypt vault data using the Web Crypto API (AES-256-GCM) and store it locally with IndexedDB.

Reference Password Manager (Avast) Interface

